```
clear;
clc;
close all;
```

1. PARAMETER DEFINITIONS

```
fc_carrier = 28e9;          % Carrier Frequency (28 GHz)
Fs_in = 1000e6;             % Input Sampling Frequency (1000 MHz)

M = 4;                      % Decimation Factor
L = 2;                      % Interpolation Factor

Fs_dec = Fs_in / M;         % Decimated Sampling Frequency (250 MHz)
Fs_out = Fs_dec * L;        % Interpolated Output Sampling Frequency (500
MHz)

N_samples = 4000;
t_in = (0:N_samples-1) / Fs_in;

% Generate 5G-like Multitone Subcarrier Signal (Baseband)
f1 = 15e6;                  % 15 MHz component
f2 = 35e6;                  % 35 MHz component

x_in = sin(2*pi*f1*t_in) + 0.5*cos(2*pi*f2*t_in);
```

2. DECIMATION STAGE

```
N_fir = 60;                 % Filter Order

fc_dec = (Fs_dec/2) / (Fs_in/2);
b_dec = fir1(N_fir, fc_dec, 'low');

% Anti-Aliasing Filtering and Downsampling
x_filtered_dec = filter(b_dec, 1, x_in);
x_dec = x_filtered_dec(1:M:end);
```

3. INTERPOLATION STAGE

```
x_up = zeros(1, length(x_dec)*L);
x_up(1:L:end) = x_dec;

% Design Low-Pass Anti-Imaging FIR Filter
fc_interp = (Fs_dec/2) / (Fs_out/2);
b_interp = fir1(N_fir, fc_interp, 'low');

% Apply Anti-Imaging Filtering
x_out = L * filter(b_interp, 1, x_up);
```

4. PERFORMANCE METRICS

```
[H, f] = freqz(b_dec, 1, 2048, Fs_in);
H_mag_db = 20*log10(abs(H));

pb_idx = f <= (Fs_dec/2 * 0.8);
sb_idx = f >= (Fs_dec/2);

passband_ripple = max(H_mag_db(pb_idx)) - min(H_mag_db(pb_idx));
stopband_atten = -max(H_mag_db(sb_idx));
```

5. COMMAND WINDOW DISPLAY

```
fprintf('=========================================================\n');
fprintf(' EXPERIMENT 14: 5G mmWave MULTIRATE PROCESSING RESULTS\n');
fprintf('=========================================================\n');

fprintf('Carrier Frequency: %.2f GHz\n', fc_carrier/1e9);
fprintf('Input Sampling Rate: %.2f MHz\n', Fs_in/1e6);
fprintf('Decimation Factor (M): %d\n', M);
fprintf('Downsampled Sampling Rate: %.2f MHz\n\n', Fs_dec/1e6);

fprintf('Signal Parameters:\n');
fprintf(' - Original Sequence Length: %d samples\n', length(x_in));
fprintf(' - Downsampled Sequence Length: %d samples\n', length(x_dec));
fprintf(' - Passband Ripple: %.2f dB\n', passband_ripple);
fprintf(' - Stopband Attenuation: %.2f dB\n\n', stopband_atten);

fprintf('Status: 5G mmWave Downconversion & Multirate Filtering
Complete.\n');
fprintf('=========================================================\n');
```

```
=========================================================
 EXPERIMENT 14: 5G mmWave MULTIRATE PROCESSING RESULTS
=========================================================
Carrier Frequency: 28.00 GHz
Input Sampling Rate: 1000.00 MHz
Decimation Factor (M): 4
Downsampled Sampling Rate: 250.00 MHz

Signal Parameters:
 - Original Sequence Length: 4000 samples
 - Downsampled Sequence Length: 1000 samples
 - Passband Ripple: 0.11 dB
 - Stopband Attenuation: 6.01 dB

Status: 5G mmWave Downconversion & Multirate Filtering Complete.
=========================================================
```

6. VISUALIZATION

```
Nfft = 2048;

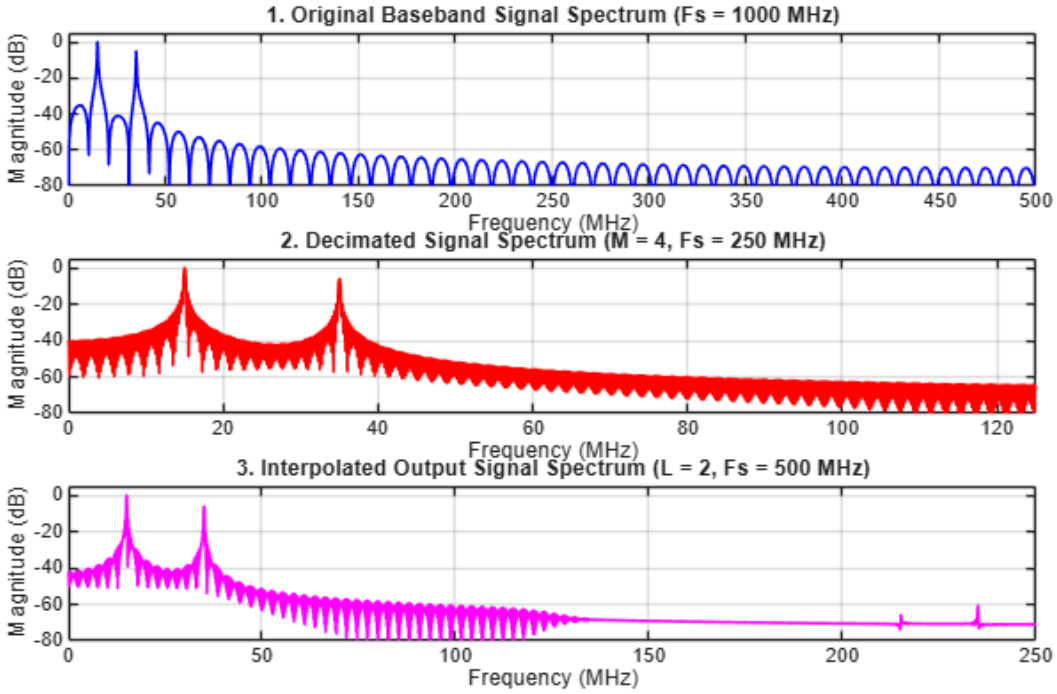
[X_in, f_in] = freqz(x_in, 1, Nfft, Fs_in);
[X_dec, f_dec] = freqz(x_dec, 1, Nfft, Fs_dec);
[X_out, f_out] = freqz(x_out, 1, Nfft, Fs_out);

figure('Name', 'Experiment 14: Multirate mmWave Processing', ...
    'NumberTitle', 'off');

% Original Signal Spectrum
subplot(3,1,1);
plot(f_in/1e6, ...
    20*log10(abs(X_in)/max(abs(X_in))), ...
    'b', 'LineWidth', 1.2);
grid on;
title('1. Original Baseband Signal Spectrum (Fs = 1000 MHz)');
xlabel('Frequency (MHz)');
ylabel('Magnitude (dB)');
xlim([0 Fs_in/2/1e6]);
ylim([-80 5]);

% Decimated Signal Spectrum
subplot(3,1,2);
plot(f_dec/1e6, ...
    20*log10(abs(X_dec)/max(abs(X_dec))), ...
    'r', 'LineWidth', 1.2);
grid on;
title(['2. Decimated Signal Spectrum (M = ', ...
    num2str(M), ', Fs = 250 MHz)']);
xlabel('Frequency (MHz)');
ylabel('Magnitude (dB)');
xlim([0 Fs_dec/2/1e6]);
ylim([-80 5]);

% Interpolated Output Spectrum
subplot(3,1,3);
plot(f_out/1e6, ...
    20*log10(abs(X_out)/max(abs(X_out))), ...
    'm', 'LineWidth', 1.2);
grid on;
title(['3. Interpolated Output Signal Spectrum (L = ', ...
    num2str(L), ', Fs = 500 MHz)']);
xlabel('Frequency (MHz)');
ylabel('Magnitude (dB)');
xlim([0 Fs_out/2/1e6]);
ylim([-80 5]);
```



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